EXERCISES

Example 1: Given the time series *Life expectancy at birth*, for the Romanian population in the period 1960-2014, estimate the trend component and the residual component using

i) Excel;
ii) EViews;
iii) R.

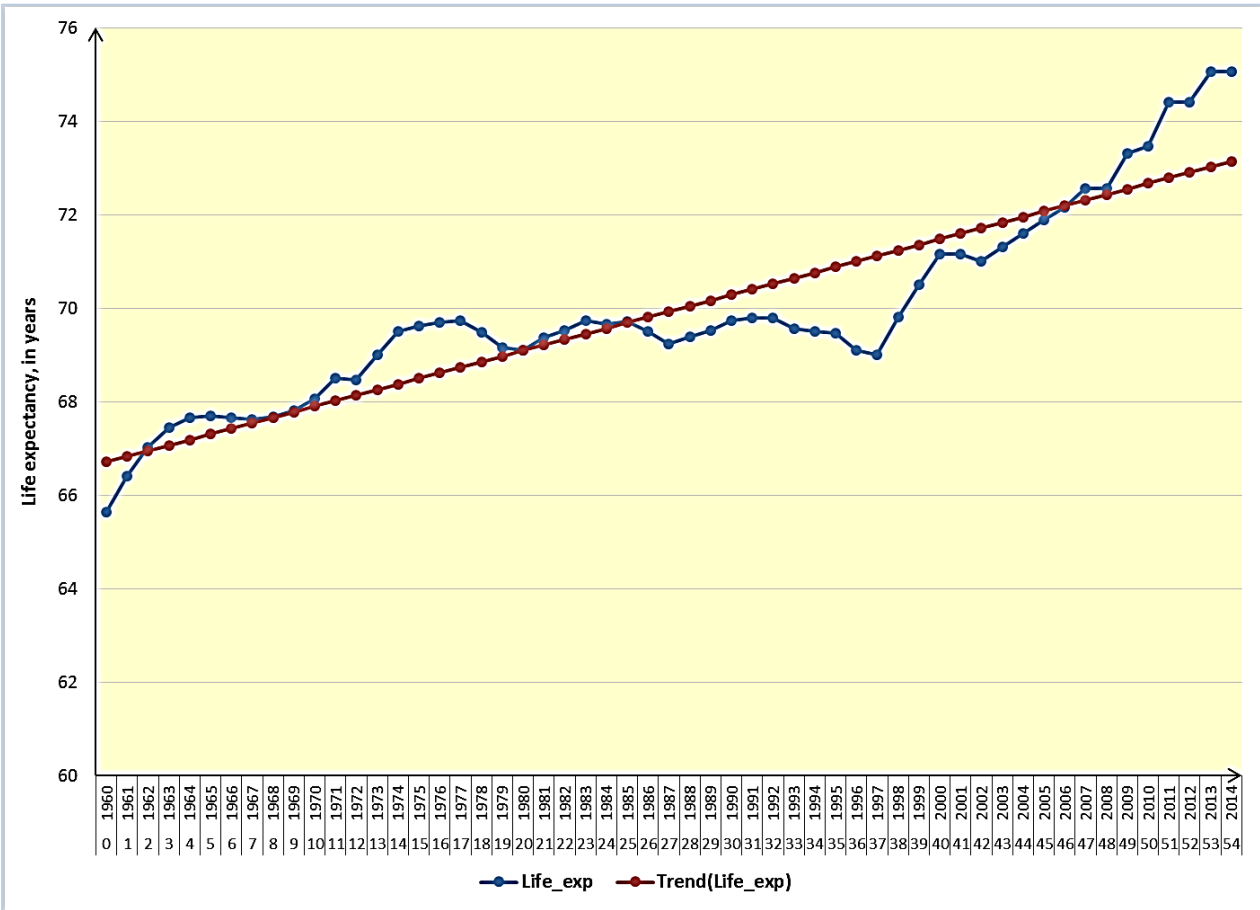
Life expectancy at birth indicates the number of years a newborn infant would live if prevailing patterns of mortality at the time of its birth were to stay the same throughout its life.

Data source: http://www.worldbank.org/

Year	Period t	Life expectancy at birth, total (years), y_t
1960	0	65.6424
1961	1	66.4159
1962	2	67.0291
1963	3	67.4411
1964	4	67.6504
1965	5	67.6979
1966	6	67.6595
1967	7	67.6269
1968	8	67.6712
1969	9	67.8144
1970	10	68.0559
1971	11	68.5041
1972	12	68.4702
1973	13	69.0056
1974	14	69.4998
1975	15	69.6139
1976	16	69.6988
1977	17	69.7420
1978	18	69.4805
1979	19	69.1532
1980	20	69.0910
1981	21	69.3683
1982	22	69.5317
1983	23	69.7263
1984	24	69.6588
1985	25	69.7068
1986	26	69.4963
1987	27	69.2268
1988	28	69.3880
1989	29	69.5307
1990	30	69.7412
1991	31	69.7844
1992	32	69.7844
1993	33	69.5634
1994	34	69.5098
1995	35	69.4561
1996	36	69.1049
1997	37	69.0049

1998	38	69.8073
1999	39	70.5122
2000	40	71.1634
2001	41	71.1610
2002	42	71.0098
2003	43	71.3098
2004	44	71.5943
2005	45	71.8789
2006	46	72.1634
2007	47	72.5659
2008	48	72.5659
2009	49	73.3098
2010	50	73.4585
2011	51	74.4098
2012	52	74.4122
2013	53	75.0634
2014	54	75.0634

Solution:

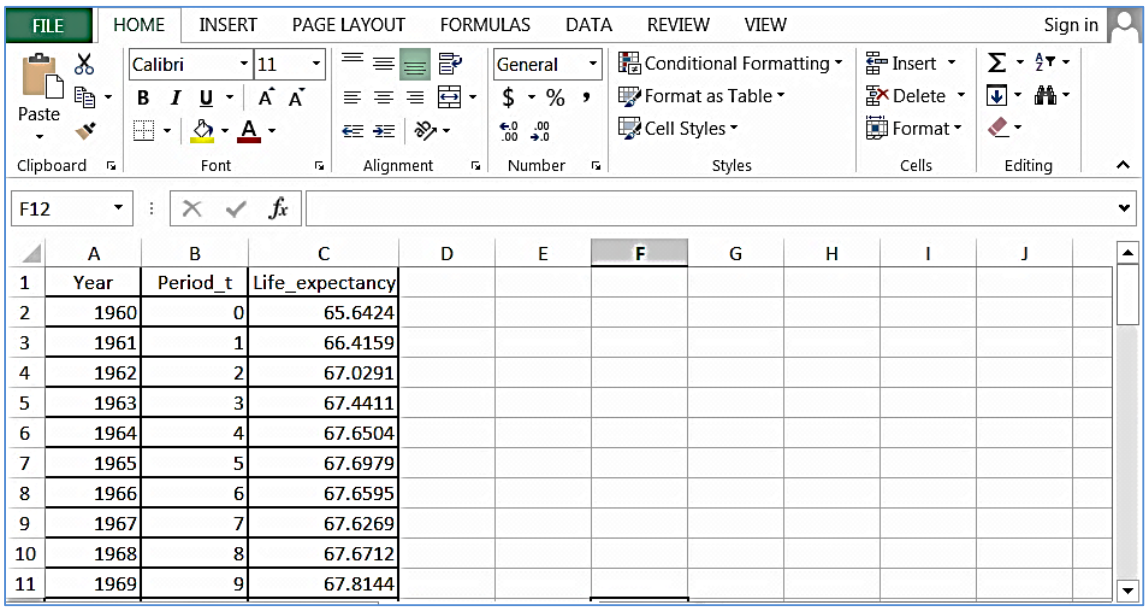


Given the time-plot of the time series **Life expectancy at birth, total (years), yearly data**, on page 2, we may assume an additive time series model without cyclical component and seasonal component, therefore the time series model is $y_t = y_{tT} + y_{tR}, t = 0,1,2,...,54$.

The time-plot suggests a linear trend: $y_{tT} = \beta_0 + \beta_1 t \Rightarrow y_t = \beta_0 + \beta_1 t + y_{tR}, t = 0,1,2,...,54$, so we have to estimate a linear regression model with *t* as the explanatory (independent) variable.

i) Excel

The excel file **Time series 1_2 2022-2023 life_expectancy Romania.xlsx** contains data on the **Year**, **Period_t** and **Life_expectancy**, as can be seen in the figure below



Year	Period_t	Life_expectancy
1960	0	65.6424
1961	1	66.4159
1962	2	67.0291
1963	3	67.4411
1964	4	67.6504
1965	5	67.6979
1966	6	67.6595
1967	7	67.6269
1968	8	67.6712
1969	9	67.8144

The time-plot, see it on page 2, suggests a linear trend:

$$y_{tT} = \beta_0 + \beta_1 t \Rightarrow y_t = \beta_0 + \beta_1 t + y_{tR},\ t = 0,1,2,\ldots,54,$$

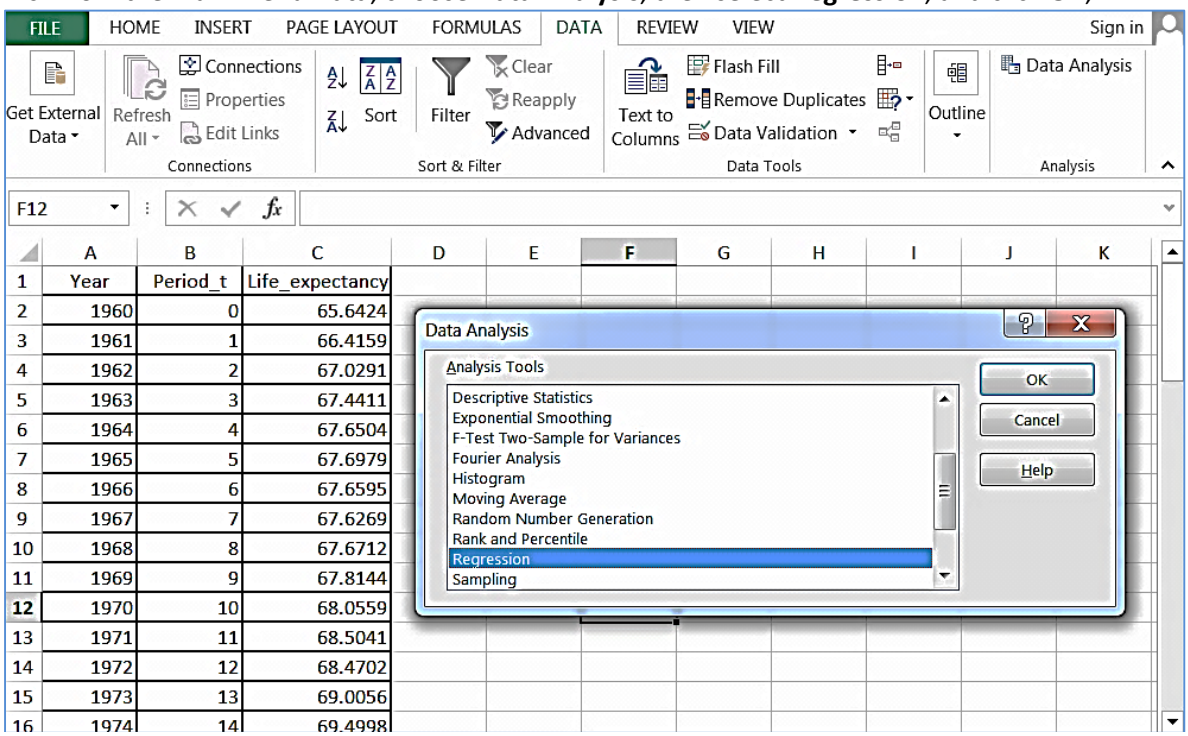
$$\text{or}\quad y_t = \beta_0 + \beta_1 t + \varepsilon_t,\ t = 0,1,2,\ldots,54,$$

so we have to estimate a simple linear regression model with

- the explanatory (or independent) variable as the time period ***t***, the observed values are given in the column denoted **Period_t** in the excel file, and
- the explained (or dependent) variable y_t, the observed values are given in the column denoted **Life_expectancy** in the excel worksheet.

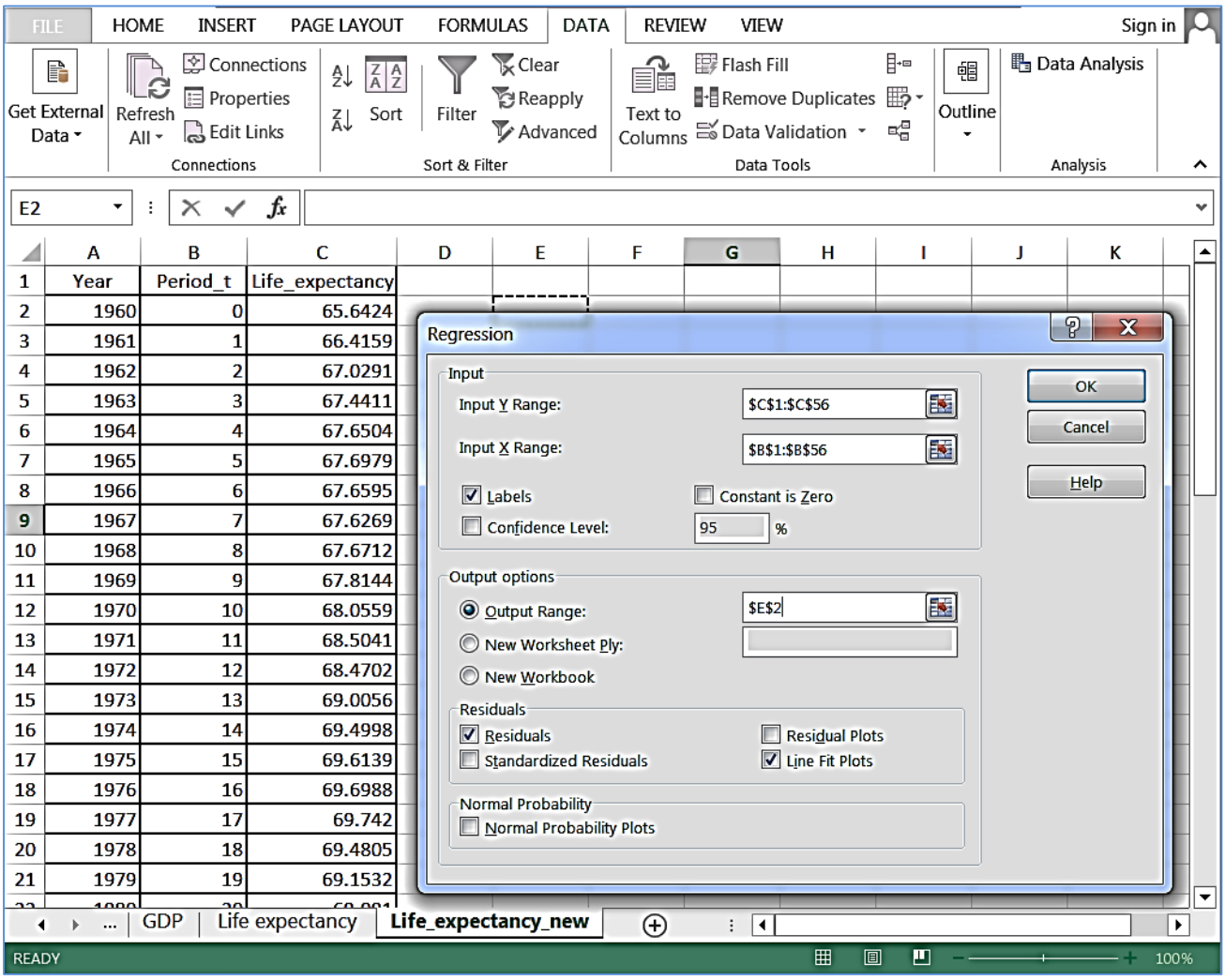
In Excel 2013, we can estimate the linear regression model in the following way:

- From the main menu **Data**, choose **Data Analysis**, then select **Regression**, and click **OK**;



In the dialog window for *Regression*:

- Select the array (range) of values of the dependent variable *Y* (**Input Y Range**), in this case the range of cells with data on **Life_expectancy**, including the name of the variable in the first cell, that is, C1:C56,
- Select the array (range) of values of the independent variable *X* (**Input X Range**), in this case the range of cells with data on **Period_t**, including the name of the variable in the first cell, that is, B1:B56,
- Check **Labels** box, as the range of sells selected for **Input Y Range** and **Input X Range** contain labels in their first entries,
- Specify the confidence level, usually **95%**,
- Specify the cell in the worksheet where to place the output (**Output Range**),
- Check **Residuals** and **Line Fit Plots**,
- Click **OK**.



Year	Period_t	Life_expectancy
1960	0	65.6424
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1962	2	67.0291
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1964	4	67.6504
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1966	6	67.6595
1967	7	67.6269
1968	8	67.6712
1969	9	67.8144
1970	10	68.0559
1971	11	68.5041
1972	12	68.4702
1973	13	69.0056
1974	14	69.4998
1975	15	69.6139
1976	16	69.6988
1977	17	69.742
1978	18	69.4805
1979	19	69.1532

The output is presented below:

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.9087
R Square	0.8258
Adjusted R Square	0.8225
Standard Error	0.8854
Observations	55

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	196.99	196.99	251.27	9.22E-22
Residual	53	41.55	0.78		
Total	54	238.54			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	$\hat{\beta}_0$ =66.7083	0.2356	283.1869	5.8E-86	66.2358	67.1808
Period t	$\hat{\beta}_1$ =0.1192	0.0075	15.8515	9.22E-22	0.1041	0.1343

The estimated trend is given by

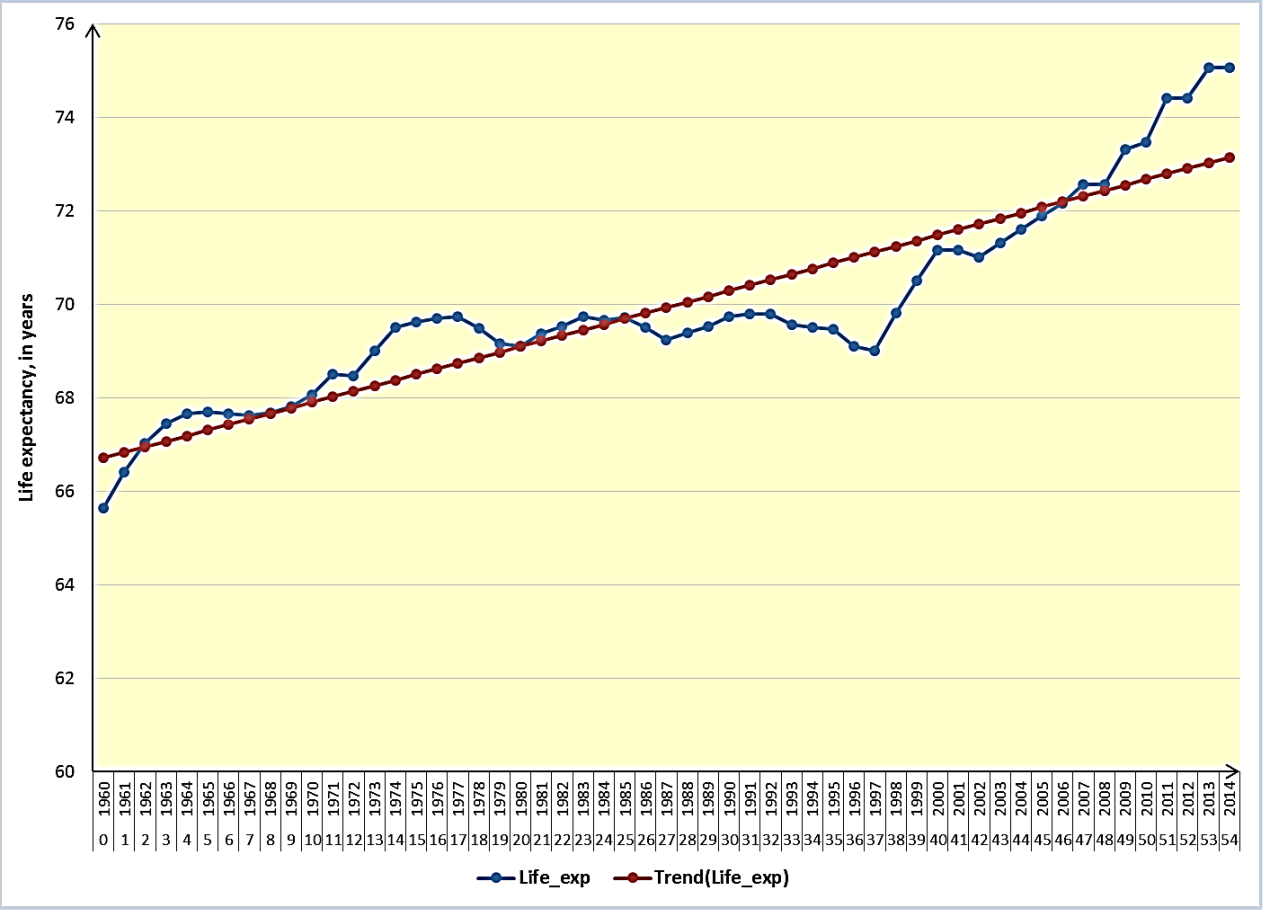
$\hat{y}_{tT} = \hat{\beta}_0 + \hat{\beta}_1 t$, that is, $\hat{y}_{tT} = 66.7083 + 0.1192\,t,\ t = 0,1,2,\ldots,54$, (*Predicted*)

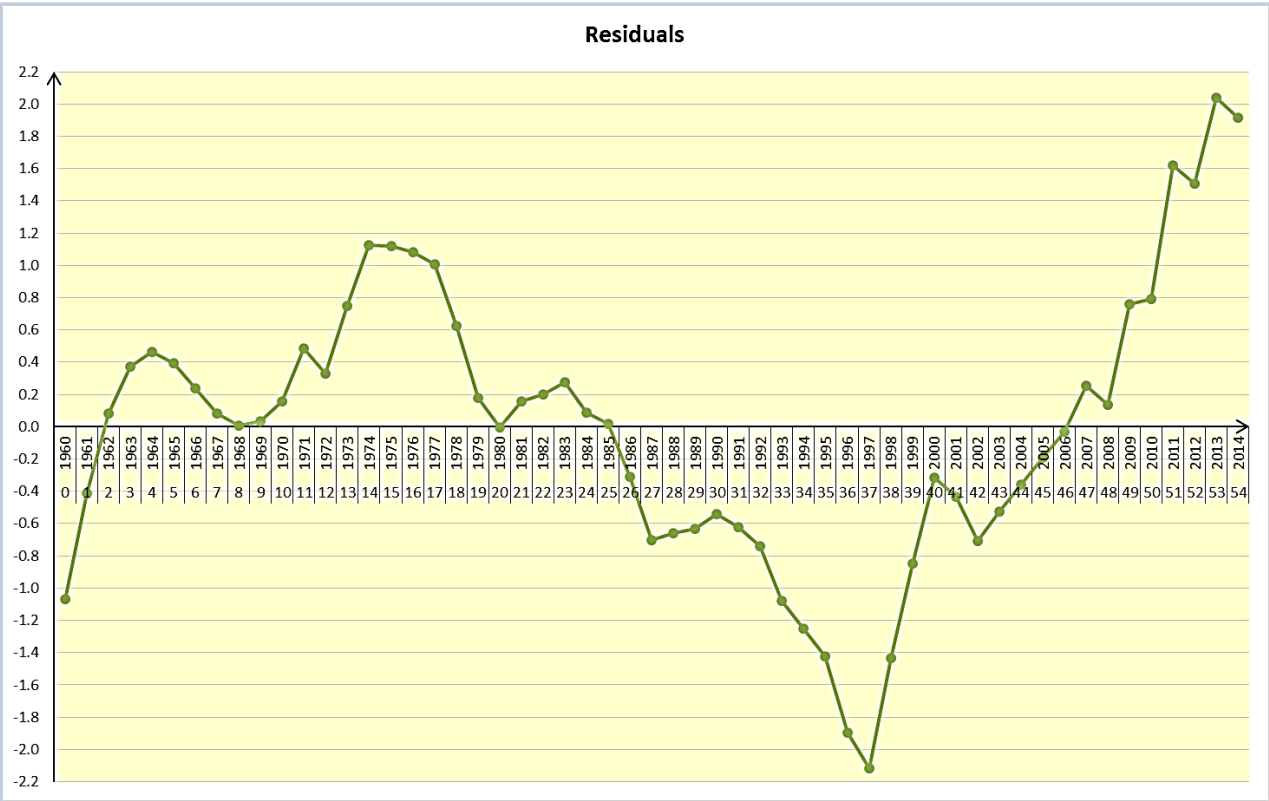
and the estimated values of the residual component (or irregular component) are given by

$\hat{\varepsilon}_t = \hat{y}_{tR} = y_t - \hat{y}_{tT},\ t = 0,1,2,\ldots,54$, (*Residuals*).

Year	Period *t*	Observed values y_t	*Predicted* $\hat{y}_{tT} = 66.7083 + 0.1192\,t$	*Residuals* $\hat{y}_{tR} = y_t - \hat{y}_{tT}$
1960	0	65.6424	66.7083	-1.0659
1961	1	66.4159	66.8275	-0.4116
1962	2	67.0291	66.9467	0.0823
1963	3	67.4411	67.0659	0.3751
1964	4	67.6504	67.1852	0.4653
1965	5	67.6979	67.3044	0.3935
1966	6	67.6595	67.4236	0.2359
1967	7	67.6269	67.5428	0.0841
1968	8	67.6712	67.6620	0.0092
1969	9	67.8144	67.7813	0.0332
1970	10	68.0559	67.9005	0.1555
1971	11	68.5041	68.0197	0.4845
1972	12	68.4702	68.1389	0.3313
1973	13	69.0056	68.2581	0.7475
1974	14	69.4998	68.3773	1.1224
1975	15	69.6139	68.4966	1.1173
1976	16	69.6988	68.6158	1.0830
1977	17	69.7420	68.7350	1.0069
1978	18	69.4805	68.8542	0.6263
1979	19	69.1532	68.9734	0.1797
1980	20	69.0910	69.0927	-0.0017
1981	21	69.3683	69.2119	0.1564
1982	22	69.5317	69.3311	0.2006
1983	23	69.7263	69.4503	0.2760
1984	24	69.6588	69.5695	0.0892
1985	25	69.7068	69.6888	0.0181
1986	26	69.4963	69.8080	-0.3116
1987	27	69.2268	69.9272	-0.7004
1988	28	69.3880	70.0464	-0.6584

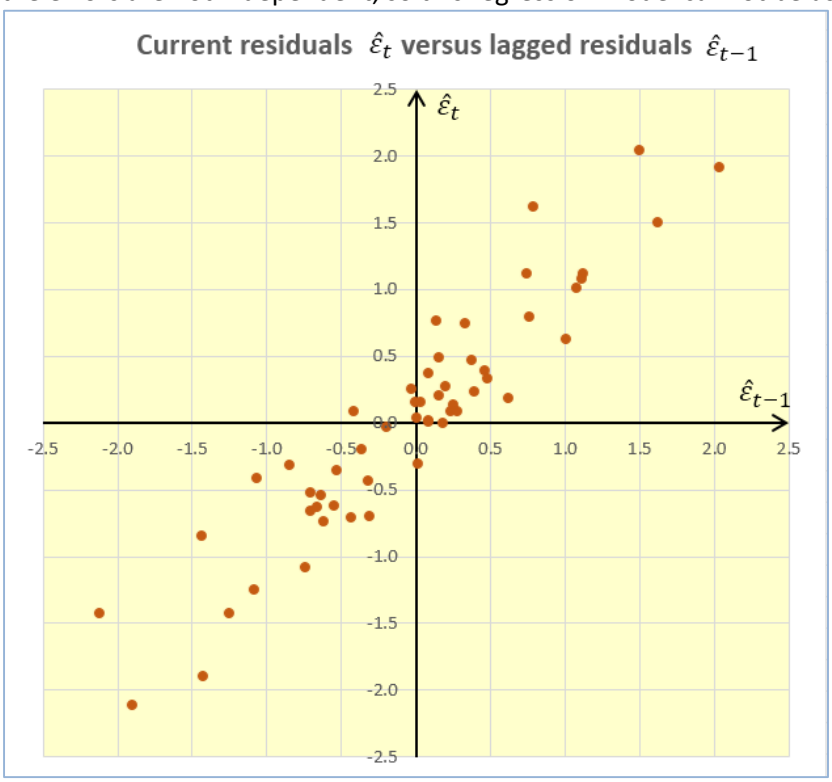
1989	29	69.5307	70.1656	-0.6349
1990	30	69.7412	70.2848	-0.5436
1991	31	69.7844	70.4041	-0.6197
1992	32	69.7844	70.5233	-0.7389
1993	33	69.5634	70.6425	-1.0791
1994	34	69.5098	70.7617	-1.2520
1995	35	69.4561	70.8809	-1.4248
1996	36	69.1049	71.0002	-1.8953
1997	37	69.0049	71.1194	-2.1145
1998	38	69.8073	71.2386	-1.4313
1999	39	70.5122	71.3578	-0.8456
2000	40	71.1634	71.4770	-0.3136
2001	41	71.1610	71.5962	-0.4353
2002	42	71.0098	71.7155	-0.7057
2003	43	71.3098	71.8347	-0.5249
2004	44	71.5943	71.9539	-0.3596
2005	45	71.8789	72.0731	-0.1942
2006	46	72.1634	72.1923	-0.0289
2007	47	72.5659	72.3116	0.2543
2008	48	72.5659	72.4308	0.1351
2009	49	73.3098	72.5500	0.7598
2010	50	73.4585	72.6692	0.7893
2011	51	74.4098	72.7884	1.6213
2012	52	74.4122	72.9077	1.5045
2013	53	75.0634	73.0269	2.0365
2014	54	75.0634	73.1461	1.9173





Does this regression model make sense?
Life expectancy can be explained only by the year someone was born?

Can it be used for prediction? If not, explain why.
As it can be seen from the plot below, there is a strong direct linear correlation between the lagged residuals $\hat{\varepsilon}_{t-1}$ and the current residuals $\hat{\varepsilon}_t$, so the errors exhibit first order autocorrelation, therefore the errors are not independent, so this regression model cannot be used for predictions.



ii) Introduce the time series data and estimate a regression model in EViews:

Open EViews software.

Then, in order to create a *workfile* (a file specific to EViews), proceed as follows:

File → New → Workfile

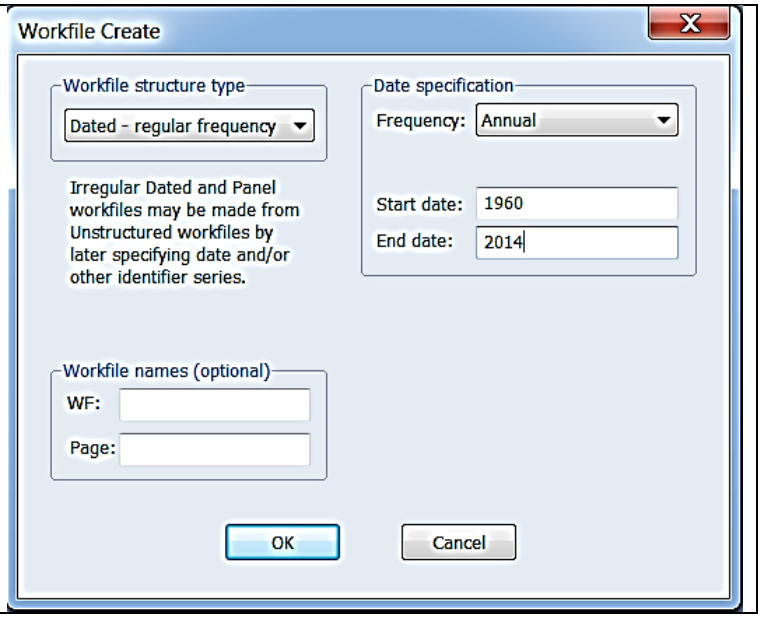
There will appear a dialog window in which:

For the field **Workfile structure type** select **Dated – regular frequency** because we will work with time series,

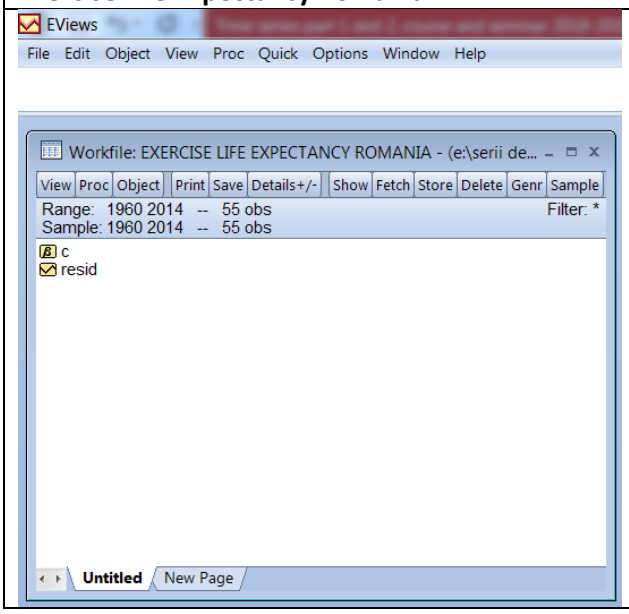
at **Data specification** field we have to specify the **Frequency**, in our case **Annual**, as we have annual data,

then you introduce the first year for **Start date**, and the last year for **End date**,

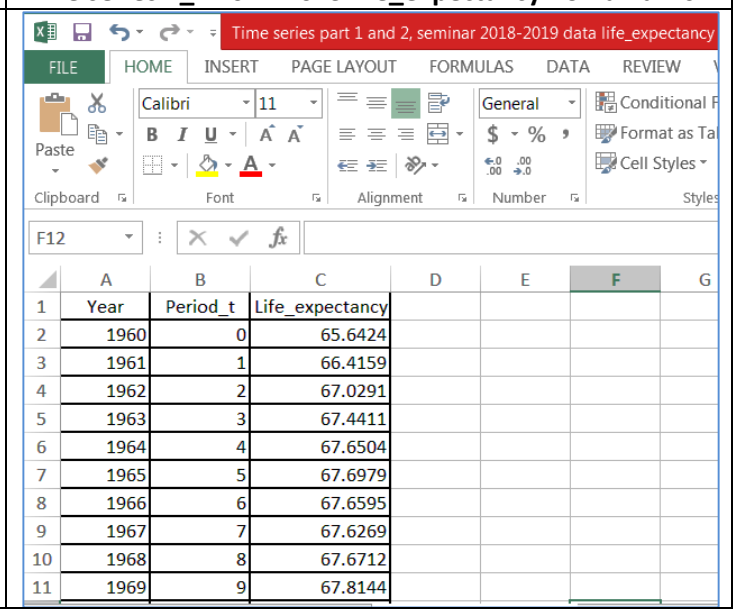
then click **OK**.



Save the workfile: **File → Save As → Exercise Life Expectancy Romania.wf1**

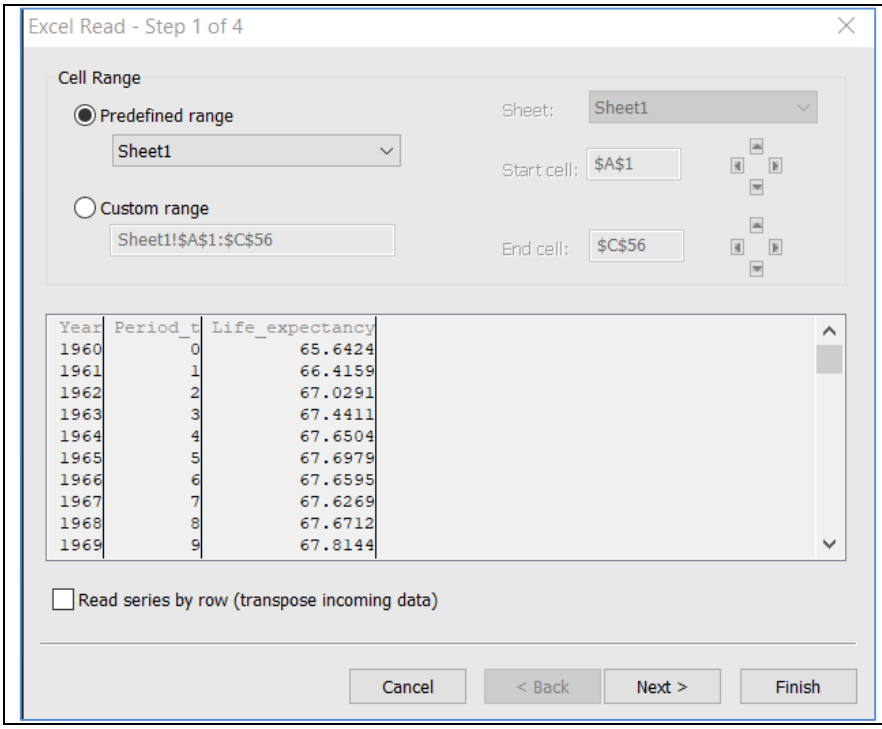


Then import the data from the Excel file **Time series 1_2 2022-2023 life_expectancy Romania.xlsx**

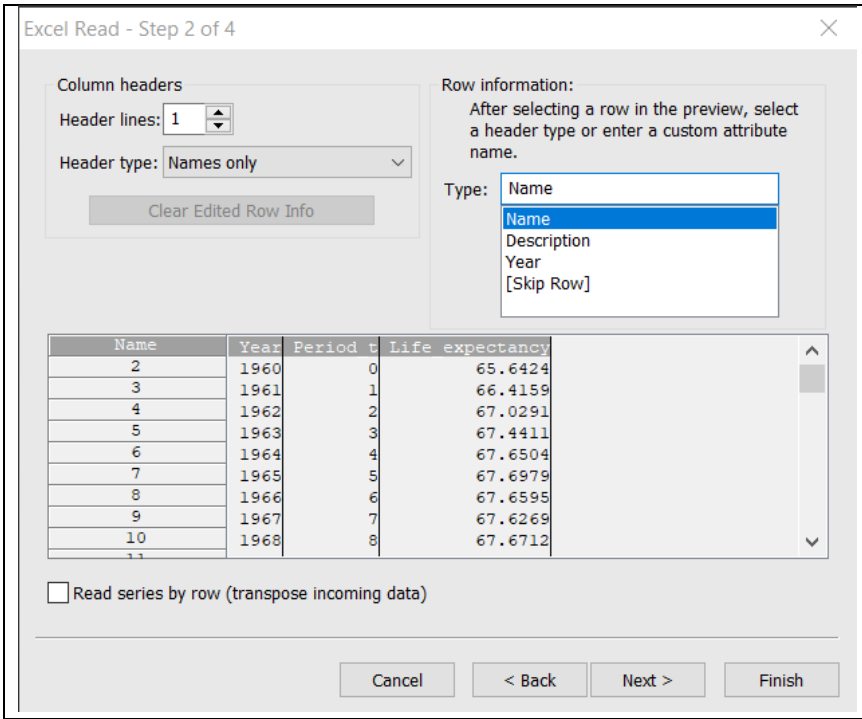


In order to **import in EViews** data from the Excel file **Time series 1_2 2022-2023 life_expectancy Romania.xlsx** we can proceed as follows:

- **File → Import → Import from File ...** ,
- Browse to select the needed excel file
- a dialog window will appear



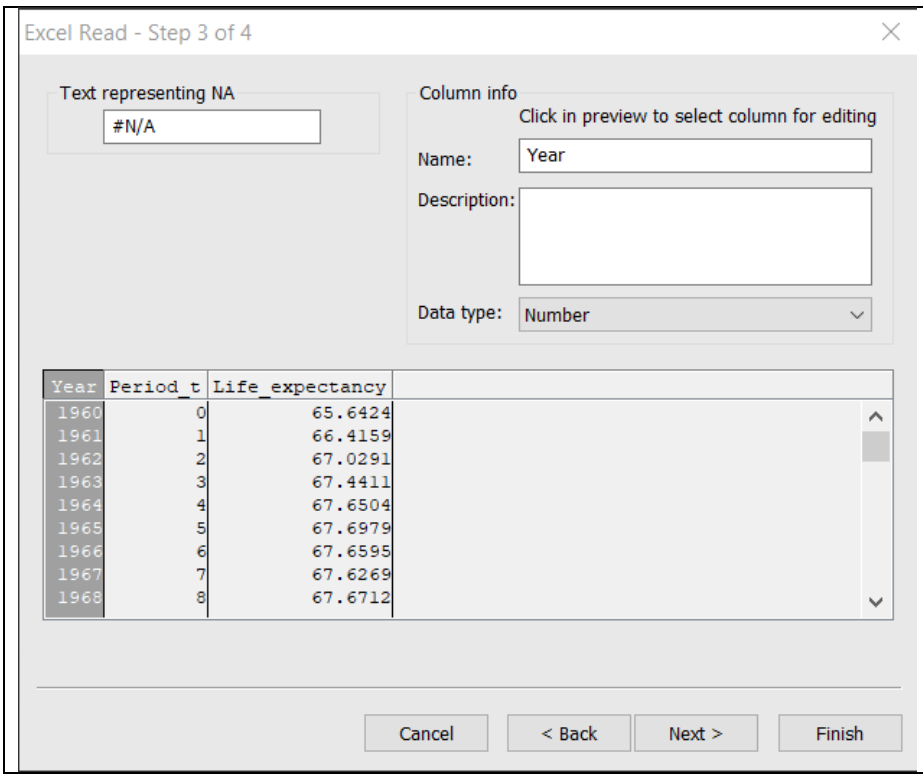
- at **Predefined range,** the worksheet denoted **Sheet1** from the excel file, it is the worksheet with the data to import
- click **Next**



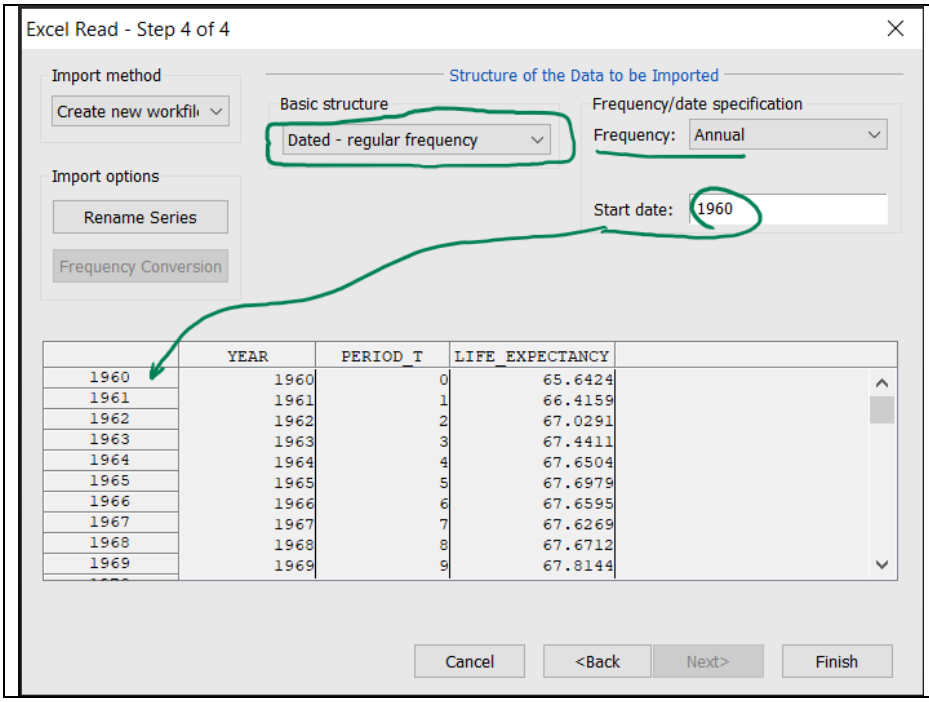
For **Column headers**,
Header lines are in the 1st line,
And
as **Header Type**, we have **Names only**

For the column with header **Year**, we specify in the field **Data Type** as being **Date**, so it will not be imported as a variable

Click on the **Finish** button



Click **Next**



In the last step 4,

at **Basic structure**, select **Dated – regular frequency**

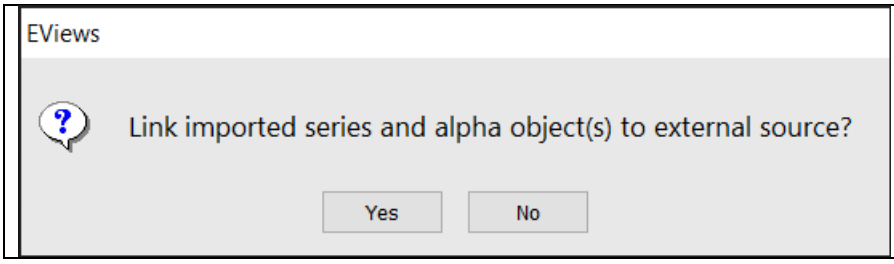
and

at **Frequency**, select **Annual**

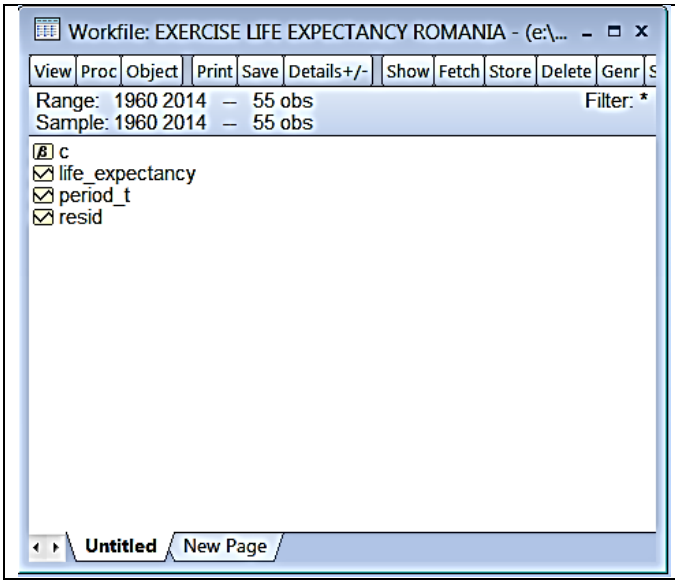
and

at **Start date**, write **1960**

and click on **Finish** button



Finally, click **Yes**

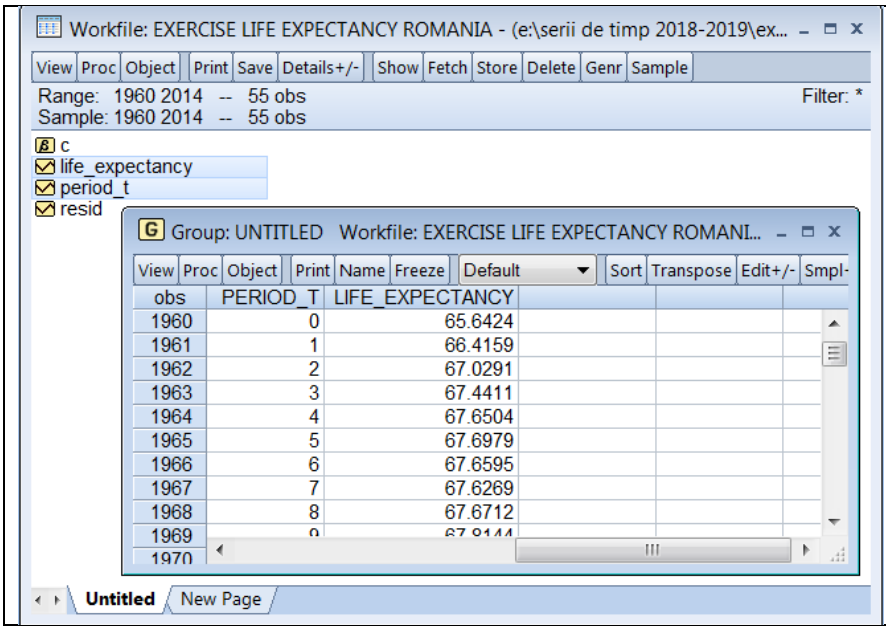


A workfile appears,
save the file with the name
Exercise Life Expectancy Romania.wf1
(in case you have the Student Lite version of EViews, you cannot save the workfile)

In the workfile there are now several objects, among them the variable **life_expectancy**

From this list of objects, select with the mouse, holding down the CTRL key, the objects **period_t** and **life_expectancy**, in this order of selection

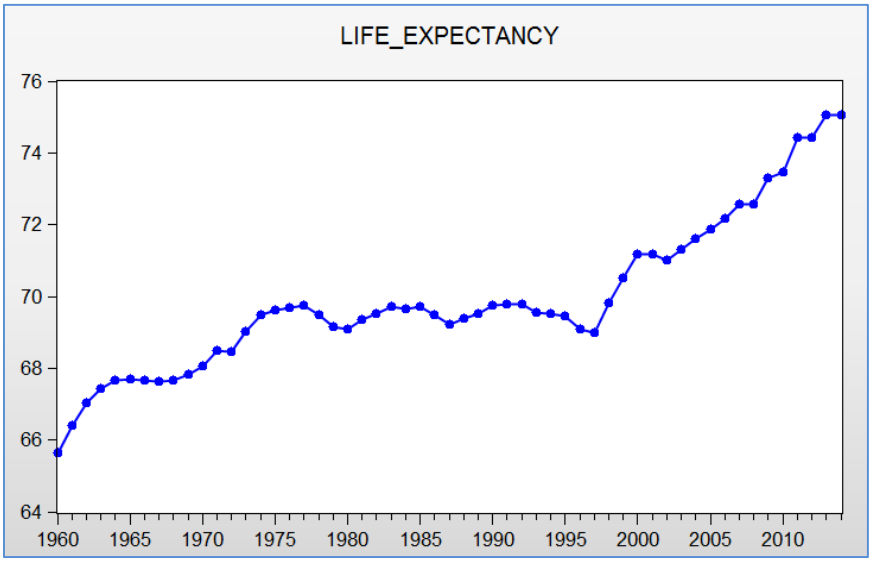
Click right **Open** → **as Group**



The data in the **Group** look like the one in the excel file, except for the variable **Year**, in this EViews workfile it is understood as the structure of the data.
The workfile was already defined as a workfile for times series.

Close the **Group** and open just the object (variable) **life_expectancy**.
Let us draw **the time series plots** for **life_expectancy**:

- in the window of the time series **life_expectancy** click on **View** → **Graph ...** → **Graph Type** (Basic type) and at **Specific** select **Line & Symbol**,



In order to estimate the linear trend of the time series **life_expectancy**, we proceed as follows:
The time-plot suggests a linear trend: $y_{tT} = \beta_0 + \beta_1 t \Rightarrow y_t = \beta_0 + \beta_1 t + y_{tR}, t = 0,1,2,...,54$,
so we have to estimate a linear regression model with *t* as the explanatory (independent) variable.

$$y_{tT} = \beta_0 + \beta_1 t \Rightarrow y_t = \beta_0 + \beta_1 t + y_{tR}, t = 0,1,2,...,54,$$

$$\text{or } y_t = \beta_0 + \beta_1 t + \varepsilon_t, t = 0,1,2,...,54,$$

so we have to estimate a simple linear regression model with

- the explained (or dependent) variable y_t, the observed values are given in the variable denoted **life_expectancy** in the workfile, and
- the explanatory (or independent) variable as the time period ***t***.

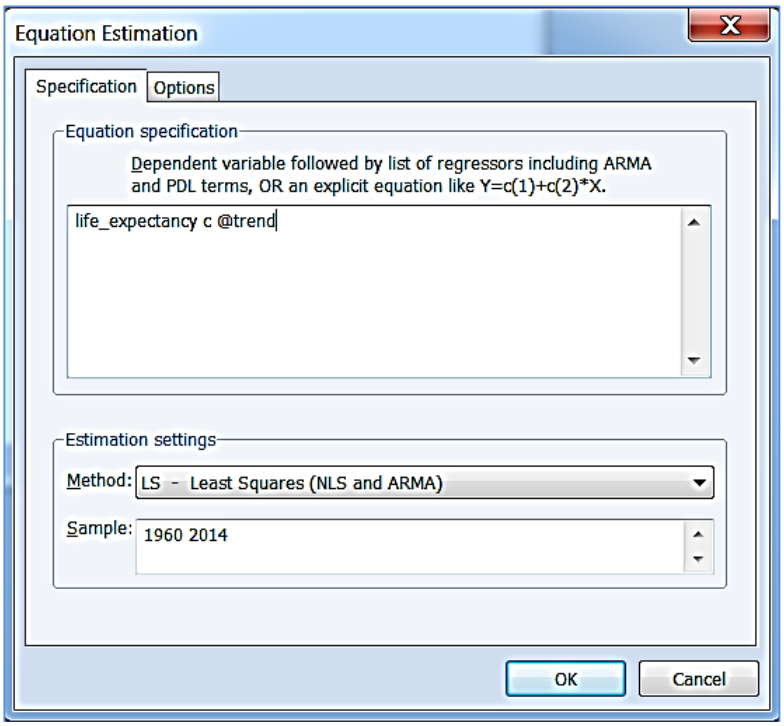
In the workfile: **Exercise Life Expectancy Romania.wf1**,

- Click on the button **Object** → **New Object ...** → **Type of object** → **Equation**, specify the name of the regression equation, that is **Name of object**, in this case **eq_trend**, would be a suggestion, for example, and then click **OK,**
- A new window, denoted **Equation Estimation**, which can be observed in the figure below, appears:
  - In the field **Equation specification**, we must specify the regression equation to estimate in the following way:

    the dependent variable, that is **life_expectancy**,
    followed by a list of regressors:
    c (which stands for the *intercept*) and
    the explanatory variable, that is the time *t*, denoted in EViews **@trend**

    life_expectancy c @trend

  - In the field **Estimation settings**, the parameters estimation method is chosen, that is **Method: LS – Least Squares** (Ordinary Least Squares Method of estimation the linear regression parameters).
  - In the **Sample** field we specify the sample of observations used in estimating the regression equation.



Equation: EQ_TREND Workfile: EXERCISE LIFE EXPECTANCY...

View | Proc | Object | Print | Name | Freeze | Estimate | Forecast | Stats | Resids

Dependent Variable: LIFE_EXPECTANCY
Method: Least Squares
Date: 02/24/19 Time: 22:25
Sample: 1960 2014
Included observations: 55

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	66.70828	0.235563	283.1864	0.0000
@TREND	0.119219	0.007521	15.85157	0.0000

R-squared	0.825814	Mean dependent var	69.92719
Adjusted R-squared	0.822527	S.D. dependent var	2.101787
S.E. of regression	0.885430	Akaike info criterion	2.630200
Sum squared resid	41.55131	Schwarz criterion	2.703194
Log likelihood	-70.33050	Hannan-Quinn criter.	2.658427
F-statistic	251.2722	Durbin-Watson stat	0.126635
Prob(F-statistic)	0.000000		

Equation: EQ_TREND Workfile: LIFE_EXPECTANCY::Unt...

View | Proc | Object | Print | Name | Freeze | Estimate | Forecast | Stats | Resids

Estimation Command:
==========
LS LIFE_EXPECTANCY C @TREND

Estimation Equation:
==========
LIFE_EXPECTANCY = C(1) + C(2)*@TREND

Substituted Coefficients:
==========
LIFE_EXPECTANCY = 66.7082829221 + 0.119218881674*@TREND

Dependent Variable: LIFE_EXPECTANCY
Method: Least Squares
Date: 02/24/19 Time: 22:25
Sample: 1960 2014
Included observations: 55

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	$\hat{\beta}_0$ =66.70828	0.235563	283.1864	0.0000
@TREND	$\hat{\beta}_1$ =0.119219	0.007521	15.85157	0.0000

R-squared	0.825814	Mean dependent var	69.92719
Adjusted R-squared	0.822527	S.D. dependent var	2.101787
S.E. of regression	0.885430	Akaike info criterion	2.630200
Sum squared resid	41.55131	Schwarz criterion	2.703194
Log likelihood	-70.33050	Hannan-Quinn criter.	2.658427
F-statistic	251.2722	Durbin-Watson stat	0.126635
Prob(F-statistic)	0.000000		

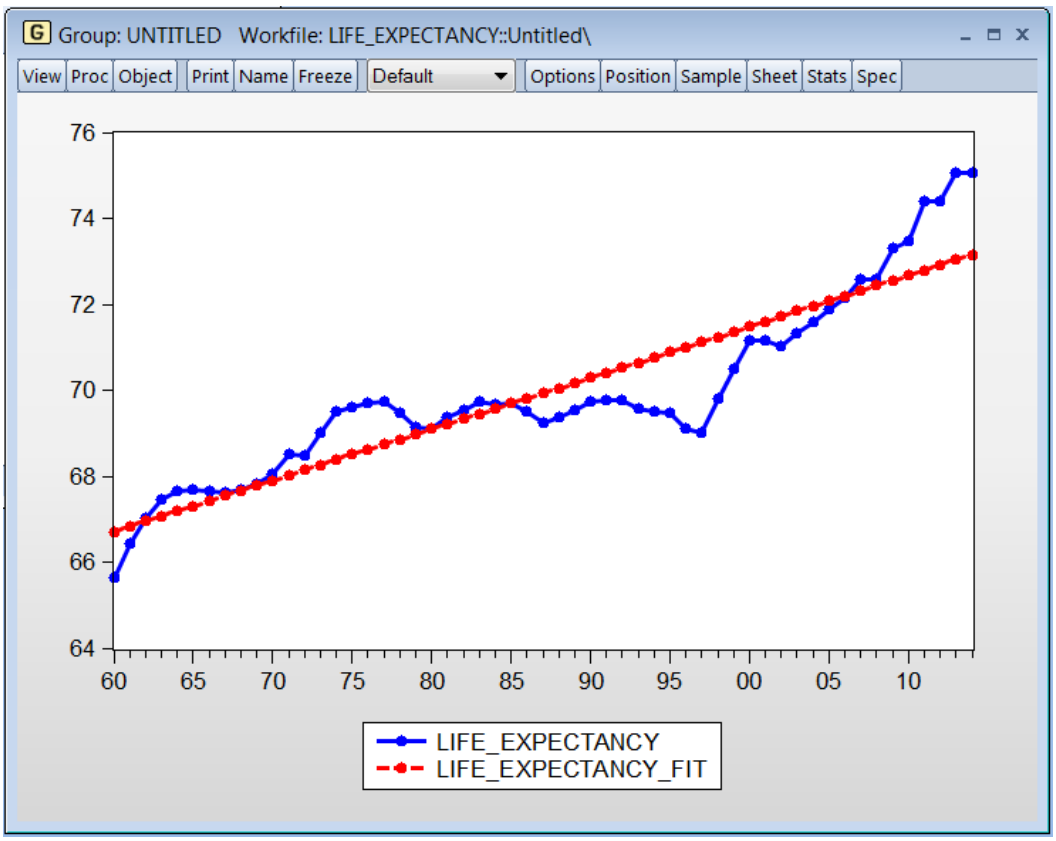
$$\hat{y}_{tT} = 66.7083 + 0.1192\,t,\ t = 0,1,2,\ldots,54$$

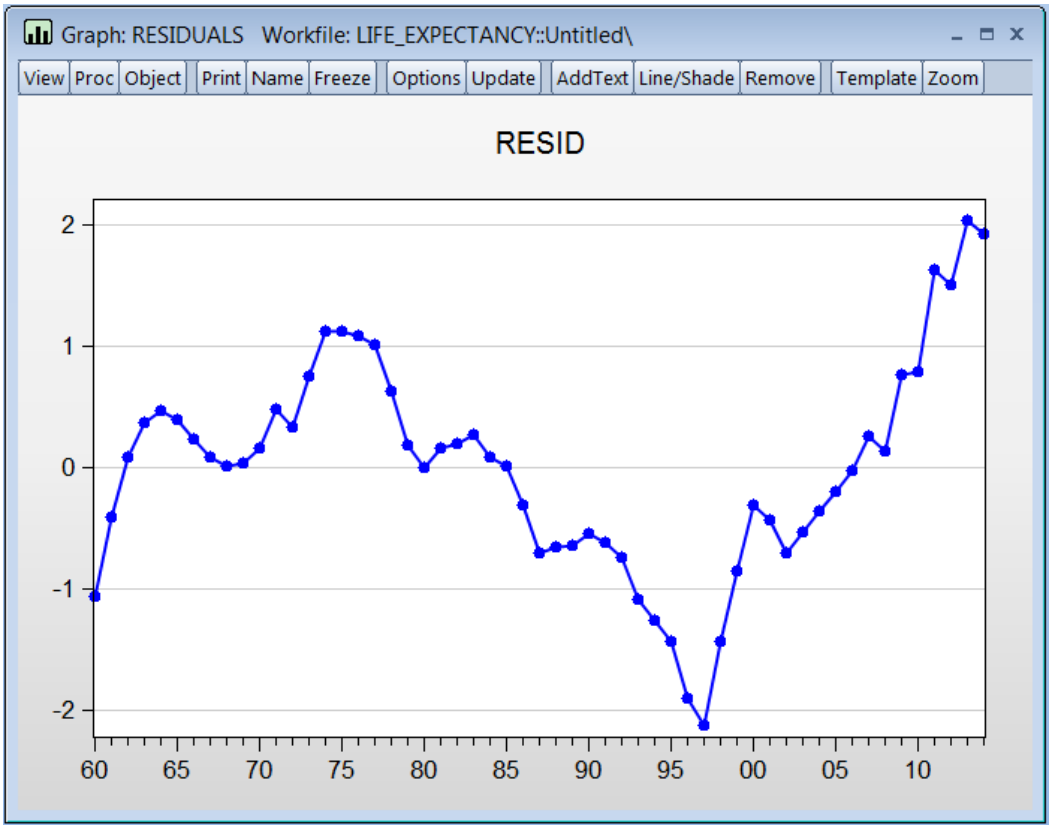
and $\hat{y}_{tR} = \hat{\varepsilon}_t = y_t - \hat{y}_{tT},\ t = 0,1,2,\ldots,54$

Equation: EQ_TREND Workfile: LIFE_EXPECTANCY::Unti...

View | Proc | Object | Print | Name | Freeze | Estimate | Forecast | Stats | Resids

obs	Actual	Fitted	Residual	Residual Plot
1960	65.6424	66.7083	-1.06588	
1961	66.4159	66.8275	-0.41160	
1962	67.0291	66.9467	0.08238	
1963	67.4411	67.0659	0.37516	
1964	67.6504	67.1852	0.46524	
1965	67.6979	67.3044	0.39352	
1966	67.6595	67.4236	0.23590	
1967	67.6269	67.5428	0.08408	
1968	67.6712	67.6620	0.00917	
1969	67.8144	67.7813	0.03315	
1970	68.0559	67.9005	0.15543	
1971				





Are the residuals uncorrelated? Comment upon the value of Durbin-Watson statistic from the output. What can you conclude?

Can this regression model be used in forecasting the life expectancy for future periods of time? If not, explain why.

iii) Use R to import the data and estimate the trend

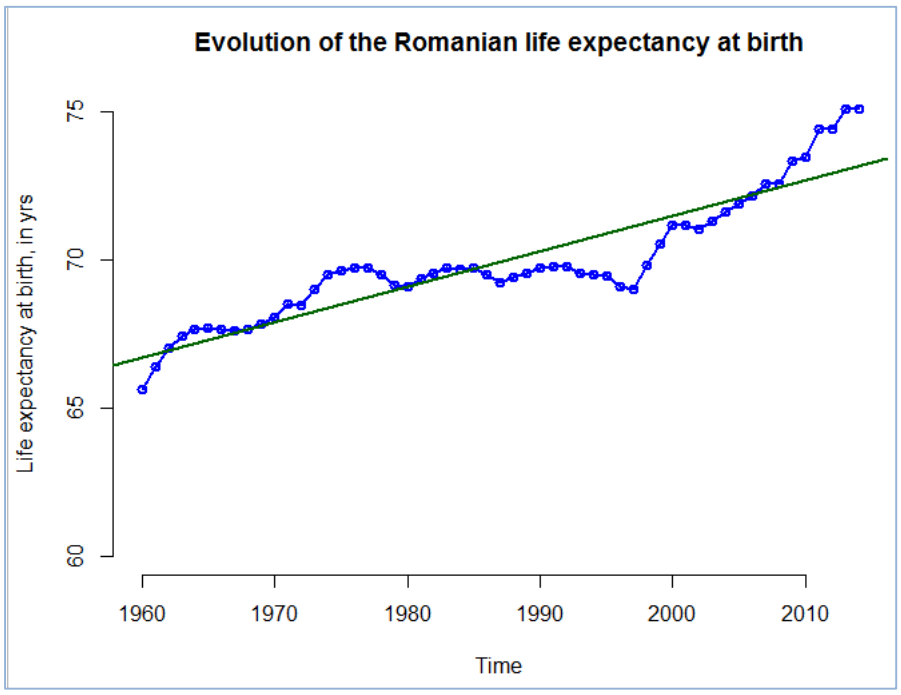
The R script is denoted **Time series part 1_2 2022-2023 script linear trend.R**

```
### Time series 1_2 2022-2023, life expectancy Romania  ###
### Estimation of a linear trend for a time series      ###
### Conf. univ. dr. Mihaela Covrig                      ###

# A. Import the data
# import into a data frame named 'example1'
# the data on life_expectancy and year
# from the file 'Time series part 1_2 2022-2023 life_expectancy Romania.csv',
# the csv file contains Year and Life_expectancy
View(example1)

# the structure of the data frame example1
str(example1)

# B. Create time series objects in the data frame
# define the variable ts_Life_expectancy as a time series object in R with data recorded yearly
?ts # help on the time series objects in R
# ts(observed values, start, frequency)
ts_Life_expectancy <- ts(example1$Life_expectancy, # the vector with the observed values of the time series
                         start=1960, # the time of the first observation
                         frequency = 1) # the number of observations per unit of time, here 1, as data is recorded yearly
ts_Life_expectancy #prints out the object
```

```
> ts_Life_expectancy #prints out the object
Time Series:
Start = 1960
End = 2014
Frequency = 1
 [1] 65.6424 66.4159 67.0291 67.4411 67.6504 67.6979 67.6595 67.6269 67.6712 67.8144 68.0559 68.5041 68.4702 69.0056
[15] 69.4998 69.6139 69.6988 69.7420 69.4805 69.1532 69.0910 69.3683 69.5317 69.7263 69.6588 69.7068 69.4963 69.2268
[29] 69.3880 69.5307 69.7412 69.7844 69.7844 69.5634 69.5098 69.4561 69.1049 69.0049 69.8073 70.5122 71.1634 71.1610
[43] 71.0098 71.3098 71.5943 71.8789 72.1634 72.5659 72.5659 73.3098 73.4585 74.4098 74.4122 75.0634 75.0634
```

```
# the type of the object ts_Life_expectancy
class(ts_Life_expectancy) # object of type "ts", that is, time series object

start(ts_Life_expectancy) # first moment of the time series
end(ts_Life_expectancy) # last moment of the time series
frequency(ts_Life_expectancy) # number of observations per unit of time
time(ts_Life_expectancy) # the time moments of the time series
```

```
# C. Plot the time series
# plot the time series
?plot
plot(ts_Life_expectancy) # generates a simple plot
# let's make the plot nicer
plot(ts_Life_expectancy, # the time series to be plotted
     type = "o", # the plot displays the line and points
     col = "blue", # color of the line plot
     lwd =2, # line width in the plot
     main = "Evolution of the Romanian life expectancy at birth", # title
     ylab ="Life expectancy at birth, in yrs", # label of the vertical axis
     ylim = c(60,75), # bounds on the vertical axis
     frame = FALSE) # the plot is without frame, see it with frame=TRUE
# we introduce the linear trend
abline(lm(ts_Life_expectancy ~ time(ts_Life_expectancy)),
       col = "darkgreen",
       lwd = 2)
```



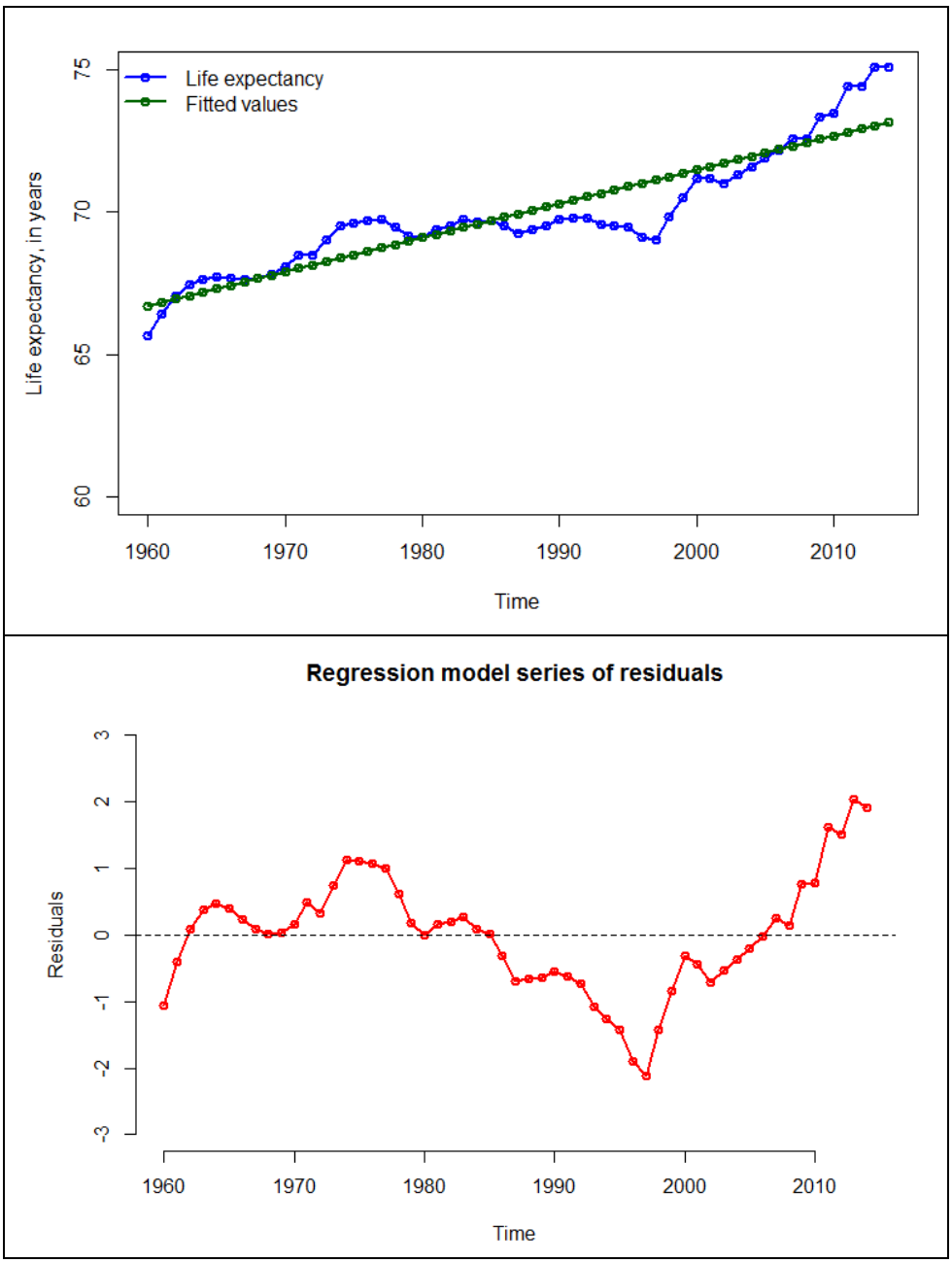

```
# D. Estimate a linear trend
# estimate a linear trend for the time series 'ts_Life_expectancy'
# a linear regression model Y = beta_0 + beta_1*X + eps
# is estimated according to the formula
# lm(Y ~ X)
# in our case the dependent variable Y is ts_Life_expectancy and X is time

# create a variable 'Period_t' with consecutive natural numbers from 0 to n-1,
# where n is the length of the time series
n <- length(ts_Life_expectancy)
n # 55 years

first_time_period <- 0
first_time_period

last_time_period <- n-1
last_time_period

?seq() # sequence generation function
Period_t <- seq(from = first_time_period,
                to = last_time_period,
                by = 1)
Period_t
head(Period_t) # first 6 values
tail(Period_t) # last 6 values
```

```
# estimate a linear trend using the linear regression model
# ts_Life_expectancy = beta_0 + beta_1*Period_t + epsilon
trend_model1 <- lm(ts_Life_expectancy ~ Period_t)
summary(trend_model1) # what do you observe about the intercept and slope
estimates?
```

```
> trend_model1 <- lm(ts_Life_expectancy ~ Period_t)
> summary(trend_model1) # what do you observe about the intercept and slope estimates?

Call:
lm(formula = ts_Life_expectancy ~ Period_t)

Residuals:
     Min       1Q   Median       3Q      Max 
-2.11448 -0.58166  0.08238  0.42938  2.03652 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 66.708283   0.235563  283.19   <2e-16 ***
Period_t     0.119219   0.007521   15.85   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.8854 on 53 degrees of freedom
Multiple R-squared:  0.8258,	Adjusted R-squared:  0.8225 
F-statistic: 251.3 on 1 and 53 DF,  p-value: < 2.2e-16
```

```
# the fitted or predicted values of the dependent variable Life_expectancy_ts
fitted_Life_expectancy <- trend_model1$fitted.values
fitted_Life_expectancy
class(fitted_Life_expectancy) # numeric vector

# the residuals of the regression model trend_model1
resid_model1 <- trend_model1$residuals
resid_model1
class(resid_model1) # a numeric vector
hist(resid_model1) # residuals histogram
summary(resid_model1) # few summary statistics of residuals

# transform the fitted values and the residuals into a time series object
ts_fitted_Life_expectancy <- ts(fitted_Life_expectancy,
                                start=1960,
                                frequency = 1)
class(ts_fitted_Life_expectancy)
ts_resid_model1 <- ts(resid_model1,
                      start = 1960,
                      frequency = 1)
class(ts_resid_model1)

plot(ts_fitted_Life_expectancy, type = "o")
plot(ts_resid_model2, type = "o")

# plot in a single graph the initial time series and the fitted values
require(graphics)  # "graphics" is a base package
?ts.plot # plots multiple time series
ts.plot(ts_Life_expectancy,
        ts_fitted_Life_expectancy,
        ylim = c(60, 75),
        type = "o",
        lwd = 2,
        gpars=list(xlab="Time",
                   ylab="Life expectancy, in years",
                   lty=c(1,1),
                   col = c("blue", "darkgreen")))
legend('topleft', # legend position in the plot
       c("Life expectancy", "Fitted values"), # explanations
       pch = c(1, 1), # point characteristic
       lty = c(1, 1), # line type
       col = c("blue", "darkgreen"), # colors
       lwd = c(2,2), # line width
       bty = "n") # no box for the legend
```

```
plot(ts_resid_model2,
     type = "o",
     col = "red",
     lwd =2,
     main = "Regression model series of residuals",
     ylab ="Residuals",
     ylim = c(-3,3),
     frame = FALSE)
abline(h = 0,
       lty = 2)
```




```
# create a data frame
table <- data.frame(time(ts_Life_expectancy),
                    Period_t,
                    ts_Life_expectancy,
                    ts_fitted_Life_expectancy,
                    ts_resid_model1)
head(table)

# enlarge the Console window to the right
round(head(table),
      digits = 4)
round(tail(table),
      digits = 4)
```

```
# save the data from the dataframe "table" into a csv file
write.csv(table,
          file = "F:/An universitar 2022-2023/Serii de timp 2022-2023/Time series 1
2022-2023/Life expectancy linear trend.csv",
          col.names = TRUE,
          row.names = FALSE)
```